Paleomagnetism of the Katherine Group in the Mackenzie Mountains: implications for post-Grenville (Hadrynian) apparent polar wander[1]

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Paleomagnetic poles from the Upper Proterozoic Mackenzie Mountains supergroup (MMs) of northwestern Canada define an apparent polar wander path lying to the west of the Grenville Loop. This path is suggested from an analysis of the quartzitic Katherine Group, whose probable primary pole lies at the beginning of the sequence, near the younger end of the Grenville Track (0.88 Ga). The end of the apparent polar wander (APW) sequence may be defined by a primary pole from sills intruding the Tsezotene Formation below the Katherine. We relate the sills, dated at about 0.77 Ga, to the rifting event that led to "Copper cycle" and Rapitan sediments above the MMs, and we suggest that the exposed part of the MMs has an age between 0.88 and 0.77 Ga. The APW path is apparently not affected by rotations: pole evidence indicates little if any relative rotation between thrust sheets of the fold belt or between the fold belt and the craton.

Paleomagnetic analysis of the Katherine Group data, obtained by alternating field, thermal, and chemical methods, revealed three magnetizations. The probable primary remanence, K_A, carried by mainly detrital hematite grains, has a direction of D, I = 267°, +21° (N = 13 specimens, k = 33, α_{95} = 7°) and a pole at 9°N, 210°W (δ_p, δ_m = 4°, 8°). A secondary component, K_B, carried by hematite pigment, has a direction of D, I = 258°, +42° (N = 4 sites, k = 326, α_{95} = 5°) and a pole at 17°N, 196°W (δ_p, δ_m = 4°, 6°). It documents further a pervasive overprint magnetization found in most MMs rocks. A similar hematite magnetization is probably primary in the overlying Copper cycle rocks. The youngest component, K_C, is found partly in a second, probably largely post-folding pigment phase (post-Late Cretaceous or Paleocene) and has a direction of D, I = 007°, +84° (N = 9 sites, k = 77, α_{95} = 6°) and a pole at 77°N, 122°W (δ_p, δ_m = 11°, 12°).

Les pôles paléomagnétiques du supergroupe Mackenzie Mountains (MMs), d'âge protérozoïque, du nord-oeust du Canada définissent une courbe de dérive apparente des pôles passant à l'ouest de la boucle du Grenville. Cette courbe est déduite d'une analyse du groupe quartzitique Katherine, dont le pôle primaire probable est attribué au début de la séquence, correspondant approximativement à l'extrémité la plus jeune de la courbe du Grenville (0,88 Ga). L'extrémité de la courbe de la dérive apparente des pôles de la séquence peut être définie par le pôle primaire des filons-couches qui recoupent la formation Tsezotene sous-jacente au groupe Katherine. Les filons-couches datés à 0,77 Ga sont reliés à l'êvénement de rifting qui est responsable de la présence du "cycle de cuivre" et des sédiments Rapitan sus-jacents au MMs, et ils indiquent pour les alleurements du MMs un âge compris entre 0,88 et 0,77 Ga. La courbe de la dérive apparente des pôles ne semble pas avoir subi par des rotations; l'étude des pôles révèle que s'il y a eu rotation entre les nappes de charriage de la zone plissée ou entre la zone plissée et le craton, elle n'a pu être que très restreinte.

Les analyses paléomagnétiques des données du groupe Katherine, obtenues par les méthodes chimiques, thermales et de champ alternatif, indiquent trois aimantations. Une rémanence probablement primaire, K_A, portée principalement par des grains détritiques d'hematite, possède une direction de D, I = 267°, +21° (N = 13 échantillons, k = 33, α_{95} = 7°) et un pôle à 9°N, 210°O (δ_p, δ_m = 4°, 8°). La deuxième composante, K_B, portée par le pigment d'hématite, possède un direction de D, I = 258°, +42° (N = 4 emplacements, k = 326, α_{95} = 5°) et un pole à 17°N, 196°O (δ_p, δ_m = 4°, 6°). Ceci renseigne en plus sur l'aimantation de surimpression pénétrante observée dans les roches du supergroupe Mackenzie Mountains. Une aimantation semblable due à l'hématite est probablement primaire dans les roches sus-jacentes du cycle du cuivre. La composante la plus juene, K_C, n'est observée que partiellement dans un second pigment, associé probablement en majeure partie à une phase de post-déformation (post-Crétacé supérieur ou Paléocène) et dont la direction est D, I = 007°, +84° (N = 9 emplacements, k = 77, α_{95} = 6°) et un pôle à 77°N, 122°O (δ_p, δ_m = 11°, 12°).

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Introduction

This paper brings together the detailed study by one of us (JKP) of paleomagnetism of Precambrian rock units in the Mackenzie Mountains, carried out over the past 7 years, and the regional geological study of the other (JDA), carried out over the past 15. It concentrates on the Katherine Group, part of the oldest exposed succession in the region, the Mackenzie Mountains supergroup. The study forms part of a program aimed at characterizing the geological column in the region by paleomagnetism and correlating it with dated sequences elsewhere in North America through a comparison of paleomagnetic poles. When more nearly complete, the paleomagnetic record from the Mackenzie Mountains may serve in part as an important reference standard for the craton. The rocks sampled have probably not been displaced more than 20 km toward the craton during Laramide deformation (Aitken *et al.* 1982) and have not been rotated relative to one another or to the craton (this paper). The establishment of a reference standard is essential before paleomagnetism is used to help solve geologi-

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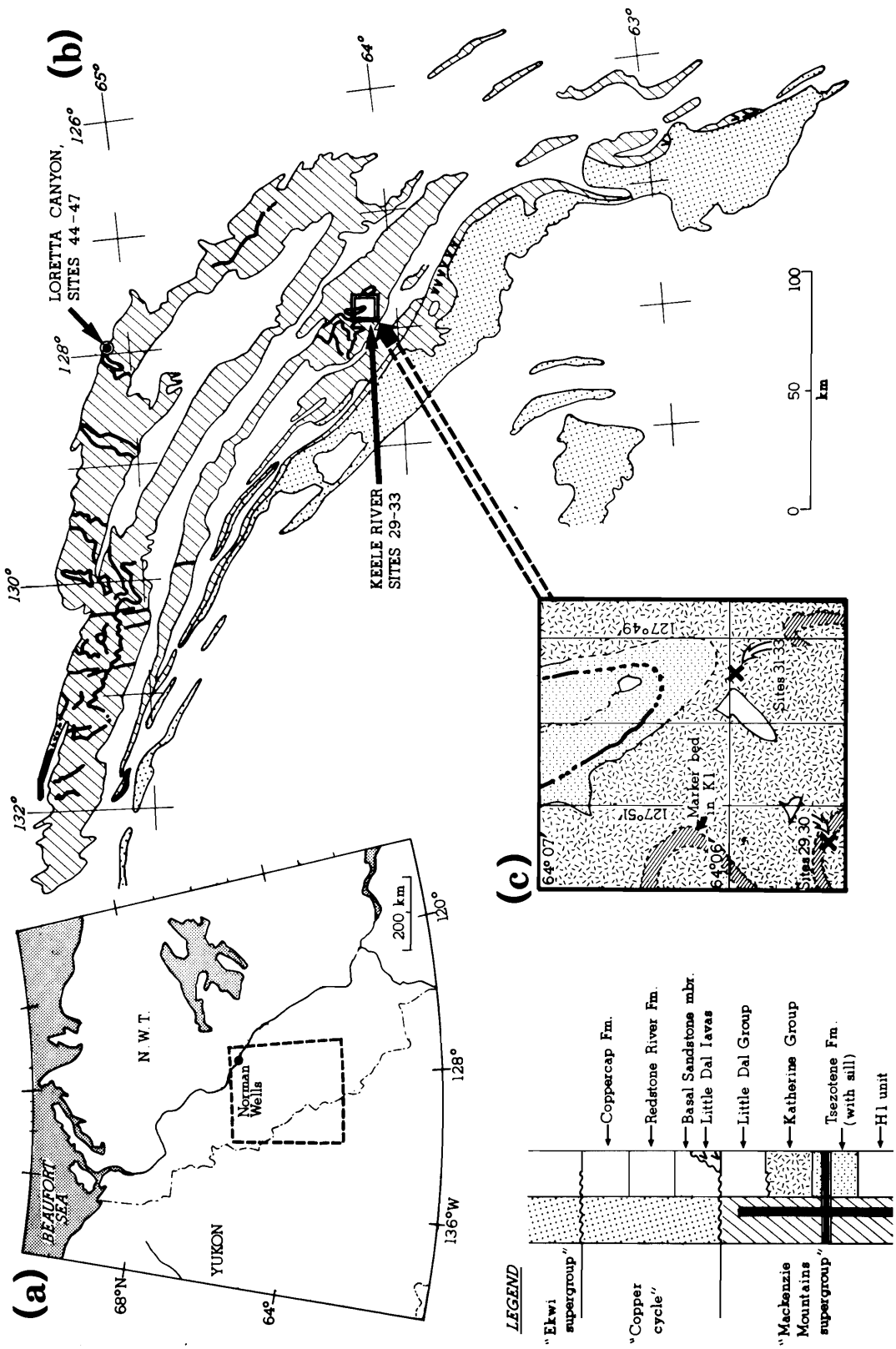


FIG. 1. (*a*) Index and (*b*, *c*) geological maps showing the sampling localities. Map (*b*) is after Armstrong *et al.* (1982). The marker bed of (*c*) is shown in Fig. 2. The left side of the legend refers to map (*b*), the right side of legend to map (*c*). The Little Dal Group is divided into six formations (see Fig. 2).

cal problems in the more complex region of the adjacent Yukon.

The Katherine Group is one of four major sedimentary units constituting the Mackenzie Mountains supergroup (Fig. 1). To date, all except the H1 unit have been sampled for paleomagnetic analysis. Results have been published for a sill intruding the Tsezotene Formation (Park 1981*b*) and for the Little Dal Group: the Mudcracked formation (Park 1984*a*), the Basinal assemblage (Park 1981*a*), and the Little Dal lavas (Morris and Park 1981; Morris and Aitken 1982). Results for the Tsezotene Formation are in press (Park and Aitken 1985), and results for the Rusty Shale formation of the upper Little Dal are still to come. Other results are available for the overlying Rapitan Group of the Ekwi (Windermere equivalent) supergroup (Morris 1977), and studies are in progress on the Basal Sandstone member and the Redstone River Formation of the intervening "Copper cycle" (Coates Lake Group of Jefferson and Ruelle 1986).

All paleomagnetic studies of Mackenzie Mountains rock units have revealed complex magnetic remanences, with from three to five or more components. It has therefore been necessary to use a variety of methods, including thermal, alternating field, and chemical treatments, to resolve the natural remanent magnetization (NRM). Because the derived paleomagnetic poles are undated, it is difficult without other constraints to relate these poles to a given apparent polar wander (APW) path.

Three major poles have been determined from the Katherine Group. The youngest appears to be associated with the Laramide Orogeny. The two older poles, like most others from the Mackenzie Mountains supergroup, lie to the west of older Belt–Purcell poles (Bressler 1981; Elston and McKee 1982). We suggest that one of the Katherine poles is primary; it would therefore be the oldest pole yet determined from the Mackenzie Mountains. The other pole, associated with hematite pigment, helps to document a pervasive magnetization, which is a chemical overprint in units of the Mackenzie Mountains supergroup but probably primary in units of the overlying Copper cycle. It may record a late Precambrian rifting event. Primary poles from the Katherine, Little Dal, and basic sills appear to document a new segment of the APW path for North America lying west of the Grenville Loop. Present evidence indicates that there has been no rotation of poles, either because of relative rotation between thrust sheets or because of relative rotation between the fold belt and the craton, to confuse this interpretation.

Geology

The Mackenzie Mountains supergroup is the older of two major, Precambrian successions exposed in the region. Aitken (1982) argued that its rocks—carbonates, mature clastics, and evaporites—and its modest thickening outward from the craton testify to a depositional environment more platformal than miogeoclinal. Correlation of the Mackenzie Mountains supergroup with the Belt–Purcell Supergroup, implying a Helikian age for the former (e.g., Gabrielse 1972; Young *et al.* 1979; Aitken *et al.* 1982), is no longer tenable in view of the fact that the Mackenzie Mountains supergroup postdates the 1.2 Ga Coppermine lavas (Young 1979), whereas folding and metamorphism of the Purcell were complete no more recently than 1.3 Ga (McMechan and Price 1982). The younger of the two major successions, the Ekwi supergroup, comprises a fining-upward megacycle initiated by a rifting event, with glaciogene rocks in its lower part (Eisbacher 1981). Basic intrusions dated at about 0.77 Ga (Armstrong *et al.* 1982) cut the Mackenzie Mountains supergroup as high as the lower Little Dal Group and yield clasts to the tillites of the basal Ekwi supergroup. Given that rocks of the Windermere Supergroup rest in two places on granitic rocks well dated at about 0.735 Ga (Evenchick *et al.* 1984), reference of the Ekwi supergroup to the Windermere Supergroup (Eisbacher 1981) is justified, both on radiometric dating and the rifting event that gave rise to both. In the Mackenzie Mountains, the two major successions are commonly, but discontinuously, separated by a subordinate, rift-related succession, the Copper cycle (Aitken 1981; Coates Lake Group of Jefferson and Ruelle 1986). Unconformable at its base (Aitken 1981) and at least locally conformable with the overlying Rapitan Group, the Copper cycle is more closely related to the Ekwi supergroup above than to the Mackenzie Mountains supergroup beneath.

Within the Mackenzie Mountains supergroup, the Katherine Group is conformable with the underlying Tsezotene Formation but at least locally disconformable with the overlying Little Dal Group. The Katherine Group is dominated by quartzites but contains minor shale and dolomite, the latter of which is commonly stromatolitic (Fig. 2). It has been divided into seven informal, numbered units of formation and member rank (Aitken *et al.* 1978). Oriented samples were collected at five sites from the lowest unit, K1, at Keele River (sites 29–33), and four sites from higher units, K4 and K5, at Loretta Canyon (sites 44–47) (Figs. 1, 2). All samples are from variously red-pigmented sandstones, ranging from very fine grained (site 30) and fine (29, 44) to medium (31–33, 45, 46) and coarse (47). The stratigraphy of the Loretta Canyon locality has been described by Aitken *et al.* (1973; units 14–12, pp. 102–103).

Methods

Block samples were oriented by sun and magnetic compasses in the field and cored in the laboratory. Cores were sliced into from one to four specimens (generally 2.5 cm diameter, 2.2 cm height).

Thermal, alternating field (AF), and chemical treatments were used singly or in combination to resolve the NRM. Thermal treatment was done on commercial (Schonstedt TSM-1) and "in-house" instruments (Roy *et al.* 1972). Specimens were heated in steps in the range 50–715°C, and their remanent magnetization was measured after each step. AF treatment was conducted on the apparatus of Roy *et al.* (1973). Steps ranged from 1 to 250 mT, with increments from 1 to 50 mT, and decay times from peak field ranged from about 40 to 300 s. Residual magnetic fields within AF and thermal treatment spaces were generally less than 2 nT; those in the Schonstedt oven cooling chamber were less than 25 nT.

For chemical treatment, axial holes were first drilled through the specimens to facilitate acid leaching (Park 1970). Then specimens were put in hydrochloric acid (6–10 N) for periods ranging from 8 h to 4 weeks. Lower acid concentrations were used in the beginning to slow down the leaching process where necessary and to prevent rapid disintegration of some specimens containing carbonate cement. At intervals, specimens were removed from acid, washed in water, dried, and measured. Leaching was halted when magnetizations became unstable or ceased to change significantly. Some specimens were leached in acid for up to 7000 h.

This general procedure was supplemented with other techniques during the long period of time over which experiments

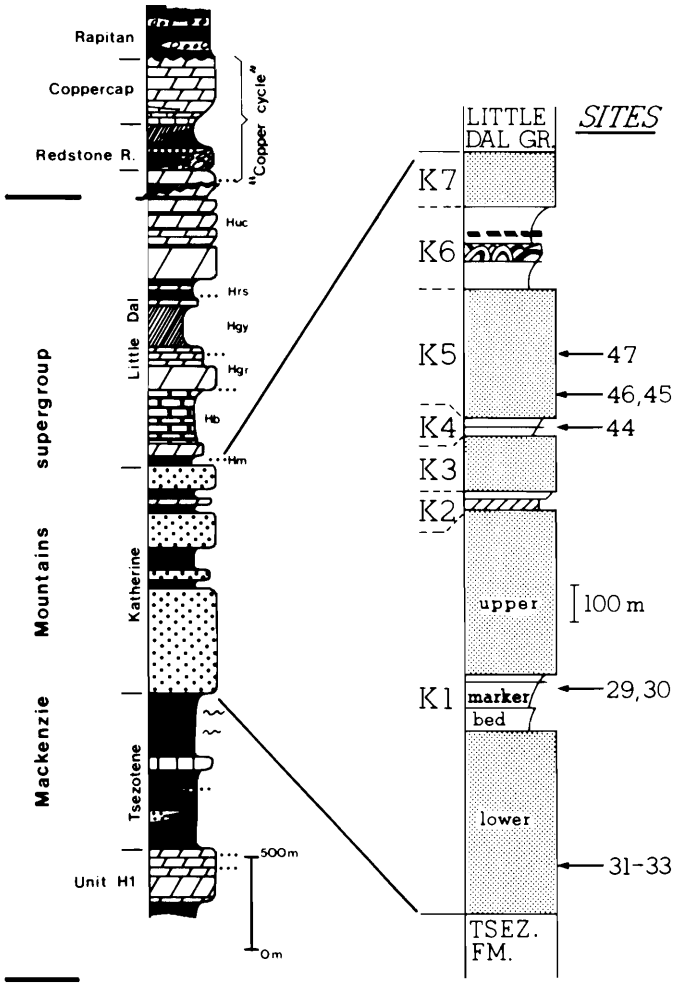


FIG. 2. Stratigraphy of the Mackenzie Mountains supergroup. The expanded portion, after Aitken *et al.* (1978), shows the major divisions of the Katherine Group and the approximate stratigraphic levels where sites were sampled. The marker bed of K1 is shown in Fig. 1*c*. Huc = Upper Carbonate formation; Hrs = Rusty Shale formation; Hgy = Gypsum formation; Hgr = Grainstone formation; Hb = Basinal–Platform assemblages; Hm = Mudcracked formation.

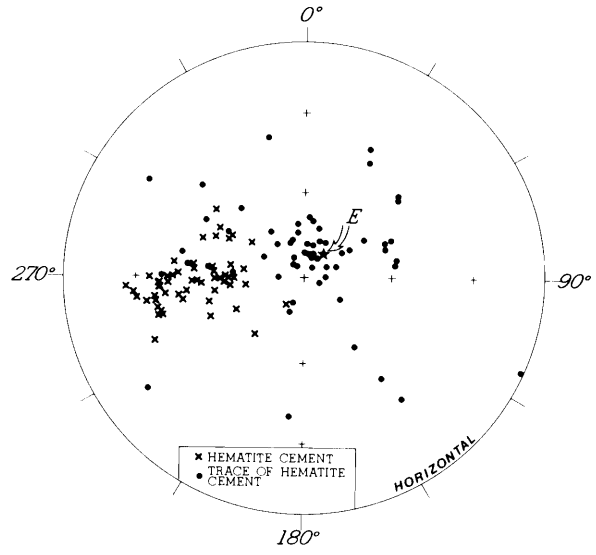


FIG. 3. The natural remanent magnetizations of specimens. Closed symbols represent downward or positive directions. E is the position of the present Earth's magnetic field in the region of sampling.

were conducted. During each of three chemical runs, specimens were routinely placed in magnetic shields after each stage of leaching for periods of sufficient length (equal to the time in acid) to passively eliminate any soft viscous remanent magnetization (VRM) acquired during leaching (Roy and Park 1974). As an experiment, during the second run, specimens were treated in a 5 mT AF after each step in order to eliminate directly soft VRM and reduce the total time of the run. However, because this method probably affected magnetic phases not leached by the acid, thus introducing effects that could not be confidently gauged, it was discontinued. During the second run, leaching was done under reduced pressure (10 kPa), which reduced leaching times by half. Steps were also taken at the time of leaching to lessen any VRM buildup. First, a Helmholtz coil was constructed about the chemical building; this reduced the Earth's magnetic field to about 12%. Second, precautions were taken to invert specimens on alternate steps, so that the buildup of soft VRM or any possible hard VRM or chemical remanent magnetization (CRM) (Henshaw and Merrill 1980) would be reduced and randomized.

Treatments were also combined. Specimens treated by AF were occasionally treated thermally, because AF either proved ineffective or introduced obvious noise components that proved to be thermally "soft". Most specimens treated chemically were additionally treated thermally or by AF. This second treatment was introduced to determine whether the magnetization of the remaining detrital grains after leaching was single or composite.

All measurements were made on computerized Schonstedt (DSM-1) or Geofyzika (JR-4) spinner magnetometers. Very weakly magnetized specimens (0.02–1 mA/m) were generally measured on the latter instrument. Specimens were transported to the magnetometers using portable magnetic shields.

Magnetic components were calculated from straight lines drawn by eye through apparent straight-line segments on vector diagrams. Many of the low intensity results below 1 mA/m were noisy, and defining components in these cases was more subjective.

Magnetic components have been combined within sites on a specimen rather than sample level. Ordinarily, samples when sufficiently dispersed within a site tend to provide an unbiased and independent estimate of the Earth's field direction. Specimens within a sample are usually too close together to provide truly independent direction determinations, but they are normally used to provide a check on the quality of data. Then, if warranted, specimen data are combined for a sample average. In this study, specimen directions are used for two main reasons: first, because the data defining the primary magnetization are limited, being confined to only 13 specimen directions from five sites; second, because the secondary magnetizations are partly chemical in nature and hence were probably acquired over a long time. Such magnetizations will be generally more dependent on time than position; therefore individual specimen directions should provide a substantial number of independent estimates of the Earth's field.

Results and analysis

Natural remanent magnetization

NRM intensities range between 0.6 and 29 mA/m and are generally quite variable within sites. The directions are posi-

TABLE 1. Characteristics of magnetic components

Treatment type	K_A (sp)	K_B (sp)	K_C (sp)
	Keele River locality		
af	?	?	10–80 mT (2)
th	650–685°C (3)	20–678°C (13)	20–550°C (4) >550°C (2), no K_B
af+th	400–710°C (2)	<700° (2)	?
ch	1435–3612 h (3)	0–4271 h (21)	0–385 h (9)
ch+th	530–630°C (1)	—	20–>550°C (11)
ch+af	?	—	2.5–>40 mT (8)
	Loretta Canyon locality		
af	?	?	1–60 mT (6)
th	?	?	20–500°C (6) 250–660°C (2), Site 47 (no K_B)
af+th	210–525°C (1)	?	<600°C (4) 400–675°C (2) Site 47
ch	24–1845 h (2)	?	24–1897 h (8)
ch+th	?	—	20–500°C (3)
ch+af	10–30 mT (1)	—	7.3–30 mT (2)

NOTES: af is alternating field; th is thermal; and ch is chemical; (sp) is the number of specimens.

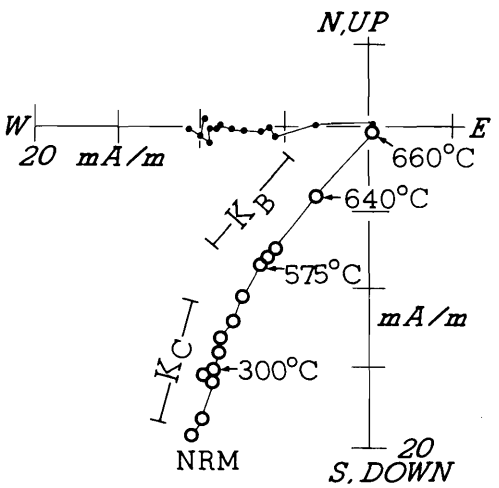


FIG. 4. Vector representation of the thermal demagnetization of specimen 171C from site 33. Closed symbols are vectors plotted on the horizontal plane (N–E); open symbols are vectors plotted on an east–west vertical plane (UP–E). The general unblocking temperature ranges at which K_B and K_C are revealed are indicated. On this and succeeding vector diagrams the vectors are plotted as if *in situ*, without bedding correction.

tive (downward directed), with inclinations ranging from steep, near the present Earth's field direction, to moderate and westerly for pigmented sites (Fig. 3).

Thermal demagnetization

Thermal treatment helped to resolve three main components (Table 1). The first component, K_C, is removed at temperatures below about 550°C and the second, K_B, in the range 550 to between 640 and 660°C (Fig. 4). K_C has a steeply inclined direction; K_B has a moderately inclined direction to the west. Both directions are positive and mimic the general NRM directions for unpigmented and pigmented specimens, respectively. A third component, K_A, is sometimes revealed thermally after AF or chemical treatment. In most specimens, the components have overlapping unblocking temperature (T_{UB}) ranges, and therefore not all of them could be resolved.

Alternating field treatment

In contrast to thermal demagnetization, AF treatment was only useful for resolving one component. This component is similar in direction to K_C and has a resistive coercive force (rcf) range from 1 to 80 mT (Table 1). The same component was revealed by AF following chemical treatment.

Chemical treatment

Chemical treatment uncovered three magnetic components similar to those revealed thermally and in the same order (Figs. 5–7). Components are sometimes mixed, that is, removed by acid at the same time, and not all were recovered from each specimen. The first, K_C, was eliminated in leaching periods from 0 to 385 h in the Keele River specimens and from 24 to 1897 h in the Loretta Canyon specimens (Table 1). The second, K_B, was removed by leaching up to 4270 h in Keele River specimens but was not found in Loretta Canyon specimens. The third component, K_A, was only resolved in five specimens (Figs. 5, 6). In the pigmented Keele River specimens, in contrast to those at Loretta Canyon, K_A was removed at higher leaching times, following the removal of K_B (Figs. 5, 6; Table 1). Occasionally, following the apparent elimination of K_A, another steeply directed component similar to K_C was partially removed (Fig. 6).

Combined treatments

Thermal treatment following AF was the most effective technique for showing the presence of K_C, but only at Loretta Canyon (Table 1). Thermal or AF treatment following chemical generally revealed a single remanence, like K_C, with T_{UB}'s usually below 550°C and rcf's usually below 40 mT (Figs. 6, 7; Table 1). Sometimes a more shallowly inclined component was removed before K_C. This component, which is usually not

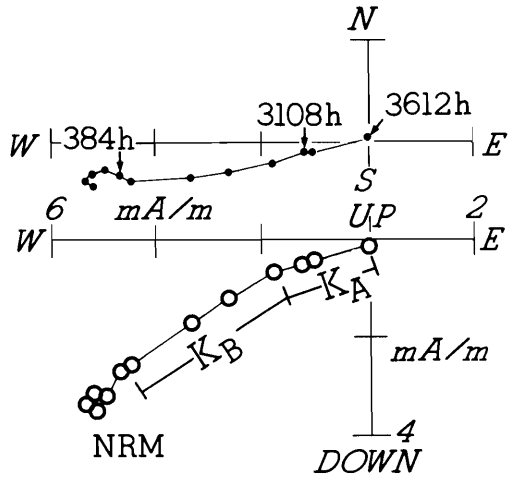


FIG. 5. Vector representation of the demagnetization of specimen 159A from site 31 during acid leaching. The presence of two magnetic components, K_A and K_B, is clearly shown. Hours are shown by h. Other details are given in Fig. 4.

clearly defined and has a variety of azimuths, was probably a soft VRM acquired during the experiments.

Directions

The use of a variety of treatments revealed three distinct, stable magnetic components, designated K_A, K_B, and K_C. These components are characterized by their directions, by their order of recovery, and to some extent, by their T_{UB}, rcf, and leaching characteristics (Figs. 4–9; Tables 1–3, 6). Directions of K_C obtained using different treatments were found to be closely similar (Fig. 8; Table 4, entries 1–6); hence, combining these directions within sites seems justified (Fig. 10; Table 2). For the same reason, the directions of K_B obtained by a variety of treatments (Fig. 9; Table 5, entries 1 and 2) are combined within sites (Fig. 10; Table 3). The direction of K_A is less well defined than those of K_B and K_C (Fig. 11; Table 6). Summaries of all three components are given in Tables 4, 5, and 7.

Carriers of the magnetization

Microscopic examination of polished thin sections established the presence of detrital and pigmental hematite at all sites except 47, which contained no pigment. The red pigment clearly occurs as two generations in the Keele site 29. No magnetite was positively identified. Some submicroscopic opaques, possibly magnetite and (or) hematite, were observed within mafic silicate particles. Minor sulphide is present in some samples.

The two generations of red pigment are clearly associated with K_C and K_B and are removed in that order at Keele River (Figs. 4, 6). Evidently, only the younger pigment containing K_C is present at Loretta Canyon (Table 1). The T_{UB}'s of K_C range up to about 675°C, the Curie point of hematite, but range below 550°C for most specimens (Table 1). In contrast, the T_{UB}'s of K_B, except in one specimen, range above 660°C. Both K_B and K_C, then, appear to reside in hematite pigment.

After the removal of K_B and K_C with the pigment phases, K_A was revealed at both Keele River (Figs. 5, 6) and Loretta Canyon. K_A must then be associated with the detrital grains, which according to the microscopic and T_{UB} evidence are hematite. Comparison of unleached and thoroughly leached

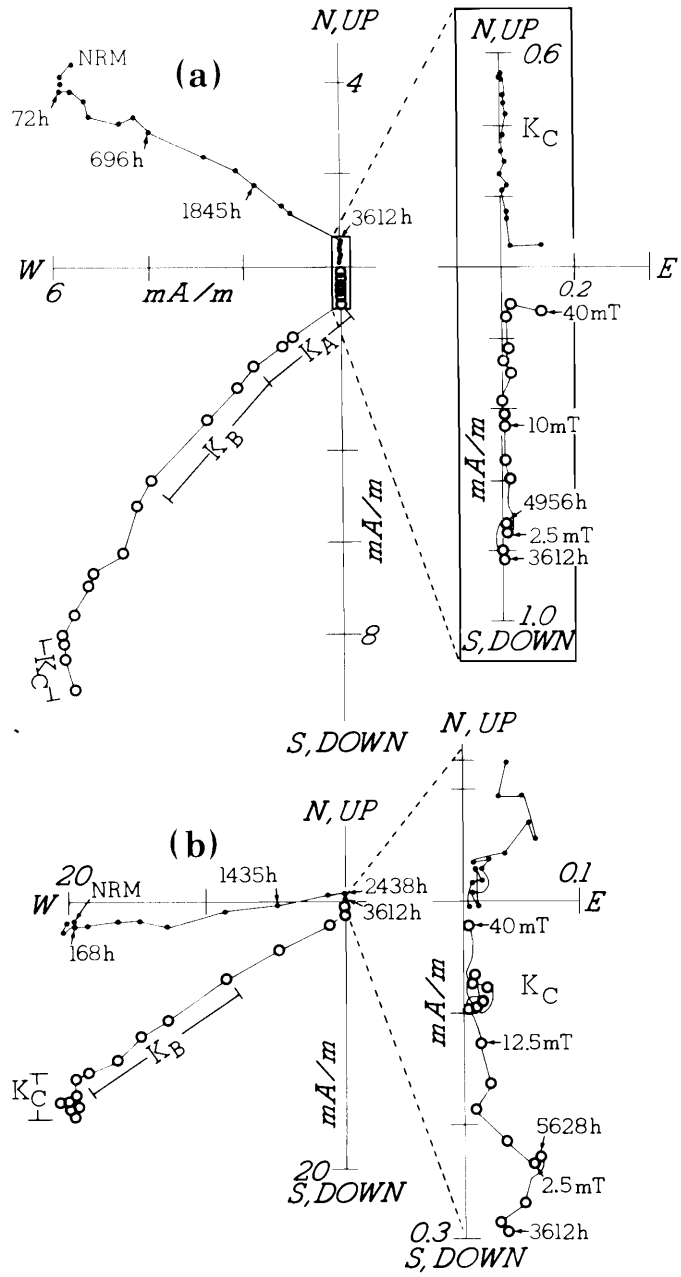


FIG. 6. Vector representation depicting the reduction of magnetization in (*a*) specimen 150A from site 30 and (*b*) specimen 164A from site 32 during a two-stage treatment of acid leaching followed by AF treatment. Other explanations are given in Figs. 4 and 5. Vectors near the origin are shown on an expanded scale on the right side, which indicates a phase of K_C remaining after leaching. The ranges over which K_A, K_B, and K_C are revealed are indicated.

specimens (e.g., site 33) revealed a distinct lack of detrital hematite in the latter, thus supporting the association of K_A with the detrital grains.

When K_A is removed, further acid leaching begins removing another steeply directed component, but it is barely affected during 2000 h of leaching (Fig. 6). This component, labelled K_C because of its direction, is fully revealed by subsequent thermal or AF treatment. It has the same range of T_{UB}'s, below 550°C, as the component revealed thermally prior to leaching, though the magnitudes are about four times smaller. Both components are probably one and the same and are not included in

TABLE 2. K_C site directions and poles

Site	Treatment	N	Directions $D°$, $I°$ (hor)	Directions $D°$, $I°$ (bdg)	R	$\alpha_{95}°$	Poles °N, °W (hor)
			Keele River locality (64.1°N, 127.8°W)				
29	th,ch,ch+th/af	5	000, +81	296, +84	4.91	12	81, 128
30	ch,af,ch+th/af	3	027, +65	016, +81	2.82	39	68, 002
31	th,ch+th/af	5	308, +78	300, +84	4.93	10	69, 186
32	th,ch,ch+th/af	5	001, +81	057, +86	4.95	9	81, 126
33	th,af,ch,ch+th/af	10	006, +85	187, +89	9.90	5	62, 129
			Loretta Canyon locality (65.1°N, 128.0°W)				
44	th,af,af+th,ch+th/af	10	228, +84	352, +75	9.71	9	57, 143
45	th,af,af+th,ch+th	8	066, +84	021, +67	7.45	13	67, 101
46	ch	3	302, +89	011, +71	3.00	3	66, 131
47	th,ch,af+th,ch+af	9	032, +81	019, +62	8.97	3	76, 083

NOTES: N is the number of specimens; D and I are the declination and inclination of the direction calculated with respect to either the present horizontal and true north (hor) or the bedding and true north (bdg); R is the vector resultant of the directions; and α_{95} is the half angle of the cone of 95% confidence about the mean direction. Poles are calculated from the average direction. See Table 1 for explanations of treatments.

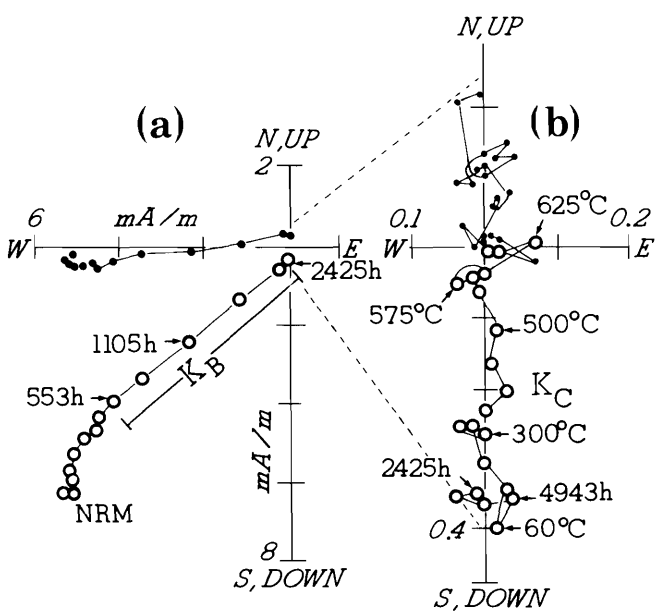


FIG. 7. Vector representation of the reduction of magnetization in specimen 161C from site 32 during progressive acid leaching followed by thermal demagnetization. The points about the origin on the left are shown on an expanded scale on the right. K_B is shown on the left, K_C on the right. Other explanations are given in Figs. 4 and 5.

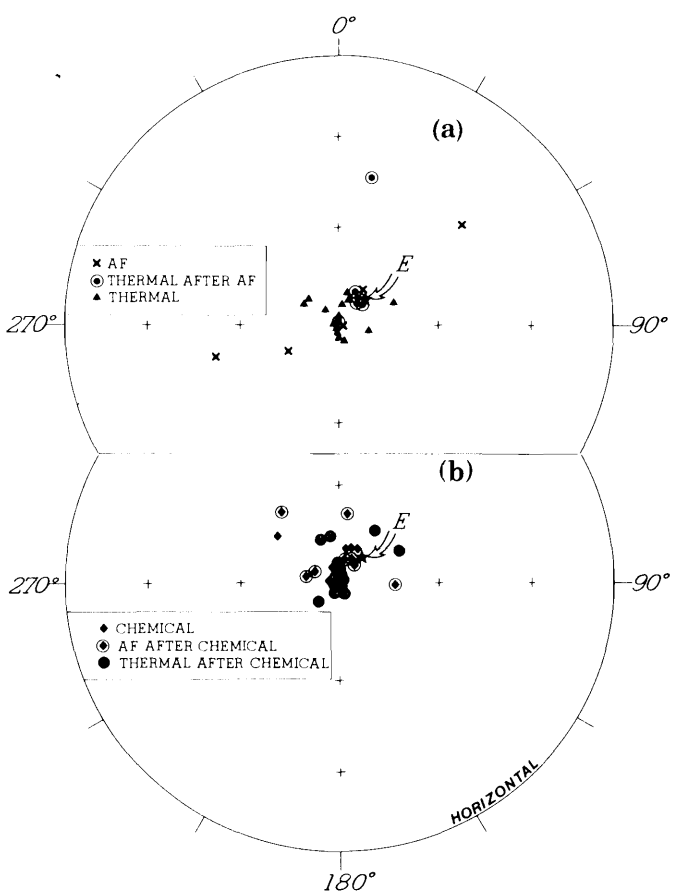


FIG. 8. K_C specimen directions. Explanations are given in Fig. 3.

the red pigment. Its rcf and T_{UB} characteristics indicate a probable magnetite carrier. We suggest that this phase of K_C resides in submicroscopic magnetite particles contained in the mafic silicates; acid would not as readily reach these particles, and so some would not reveal themselves except by thermal or AF treatment.

Tilt test

A true fold test on the directions of the three components is not possible, because sites do not occur on the opposite limbs of a fold. However, a "tilt" test can be made by comparing the scatter of directions before and after a correction for tilting of the beds (Table 8). This comparison, using the calculations for a fold test (McElhinny 1964), strongly indicates that K_B is prefolding. A test on the K_C data is equivocal, but it is significant at the 90% probability level when only Keele River data are us-

TABLE 3. K_B site directions and poles

Site	Treatment	N	Directions $D°, I°$ (hor)	Directions $D°, I°$ (bdg)	R	$\alpha_{95}°$	Poles °N, °W (bdg)
Keele River locality (64.1°N, 127.8°W)							
29	—	—	—	—	—	—	—
30	th,ch,af+th	7	274, +55	259, +45	6.87	9	19, 196
31	th,ch	8	265, +32	262, +37	7.93	6	15, 201
32	th,ch,af+th	12	261, +41	255, +42	11.89	4	16, 194
33	th,ch	9	263, +47	257, +45	8.83	8	18, 194

NOTE: See Table 1 and 2 for explanation of symbols and treatments.

TABLE 4. Summary of K_C directions and poles

Description		N	Directions $D°, I°$	R	k	$\alpha_{95}°$	Poles °N, °W	R	K	$A_{95}°$	$\delta_p°, \delta_m°$
1. af, *in situ*	sp	8	088, +88	7.09	8	21	—	—	—	—	—
2. ch, *in situ*	sp	17	354, +86	16.84	100	4	—	—	—	—	—
3. th, *in situ*	sp	14	013, +87	13.86	96	4	—	—	—	—	—
4. af+th, *in situ*	sp	6	021, +78	5.77	22	15	—	—	—	—	—
5. ch+af, *in situ*	sp	10	353, +80	9.68	28	9	—	—	—	—	—
6. ch+th, *in situ*	sp	14	009, +86	13.77	57	5	—	—	—	—	—
7. Total, *in situ*	sp	58	006, +85	56.14	31	4	74, 124	—	—	—	7, 7
8. Total, bdg	sp	58	009, +79	55.49	23	4	85, 083	—	—	—	7, 8
9. Keele, R., *in situ*	si	5	002, +80	4.94	67	10	84, 120	—	—	—	17, 18
10. Keele, R., bdg	si	5	314, +87	4.98	225	5	70, 133	—	—	—	10, 10
11. Loretta, *in situ*	si	4	038, +88	3.98	154	7	68, 121	—	—	—	15, 15
12. Loretta, bdg	si	4	013, +69	3.98	138	8	76, 343	—	—	—	11, 13
13. Total, *in situ*	si	9	007, +84	8.90	77	6	77, 122*	8.68	25	11	11, 12
14. Total, bdg	si	9	009, +79	8.85	53	7	—	—	—	—	—

NOTES: The first six entries give the number of specimens (sp) done under different treatments. The total of these is different from the totals given under entries 7 and 8, because one specimen may have direction determinations from two different treatments, whereas the total includes only one average value from each specimen. N is the number of specimens (sp) or sites (si); k is Fisher's (1953) index of precision; δ_p, and δ_m are the semi-axes of the ellipse of confidence at probability level 0.05 about the pole calculated from the average direction; R, K, and A_{95} given under "Poles" are the analogues of those parameters given under "Directions." Other symbols are explained in Tables 1 and 2. The dashed line divides the specimen averages from the site averages. Entry 13 is considered to be the best estimate of K_C.

*This pole is the average of the nine *in situ* site poles; the others are the poles determined from the average of the site directions.

ed. However, a fold test applied to data from the underlying Tsezotene Formation implies that both K_C phases are postfolding, that is, post-Paleocene (Park and Aitken 1986). There are evidently not enough data to test K_A.

Discussion

Nature of the magnetizations

K_A appears to be a depositional or detrital remanent magnetization carried mainly by hematite. One reversal was documented in site 45.

K_B is a secondary magnetization carried by hematite pigment in sites from the Keele River locality. It documents a chemical overprint magnetization, which is present in most pre-Rapitan Precambrian units of the Mackenzie Mountains (Park 1984*b*). It may record a late Precambrian rifting event (see below).

Part of K_C, also carried by pigment, appears to be mostly post-folding, having a direction whose error ($\alpha_{95}°$) includes the present Earth's field (PEF) direction in the region. Irving and McGlynn (1981) noted a similar stable, steeply directed

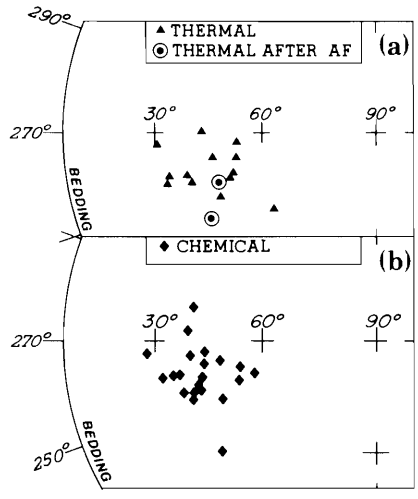


FIG. 9. K_B specimen directions. Explanations are given in Fig. 3.

TABLE 5. Summary of K_B directions and poles

Description	N	Directions D°, I°	R	k	α_{95}°	Poles °N, °W	R	K	A_{95}°	δ_p°, δ_m°
1. ch,bdg, sp	21	259, +41	20.70	66	4	—	—	—	—	—
2. th,af+th,bdg, sp	15	256, +43	14.76	59	5	—	—	—	—	—
3. *In situ*, sp	36	265, +43	35.17	42	4	—	—	—	—	—
4. Bedding, sp	36	258, +44	35.44	63	3	—	—	—	—	—
5. *In situ*, si	4	265, +44	3.95	63	12	—	—	—	—	—
6. Bedding, si	4	258, +42	3.99	326	5	17, 196	3.99	423	5	4, 6

NOTES: Symbols are explained in Tables 1, 2, and 4. Entry 4 is considered to be the best estimate of K_B.

TABLE 6. K_A site directions and poles

Site	Treatment	N	Directions D°, I° (hor)	D°, I° (bdg)	R	α_{95}°	Poles °N, °W (bdg)
			Keele River locality (64.1°N, 127.8°W)				
29,30	—	—	—	—	—	—	—
31	th,ch,af+th	7	263, +15	261, +20	6.89	8	05, 205
32	ch+th	1	286, +06	285, +11	—	—	12, 229
33	af+th	1	277, +21	275, +21	—	—	12, 217
			Loretta Canyon locality (65.1°N, 128.0°W)				
44	ch,ch+th,af+th	3	266, +27	276, +28	2.92	25	16, 217
45	ch+af	1	074, −06	077, −15	—	—	02, 203
46, 47	—	—	—	—	—	—	—

NOTE: See Tables 1 and 2 for explanation of symbols.

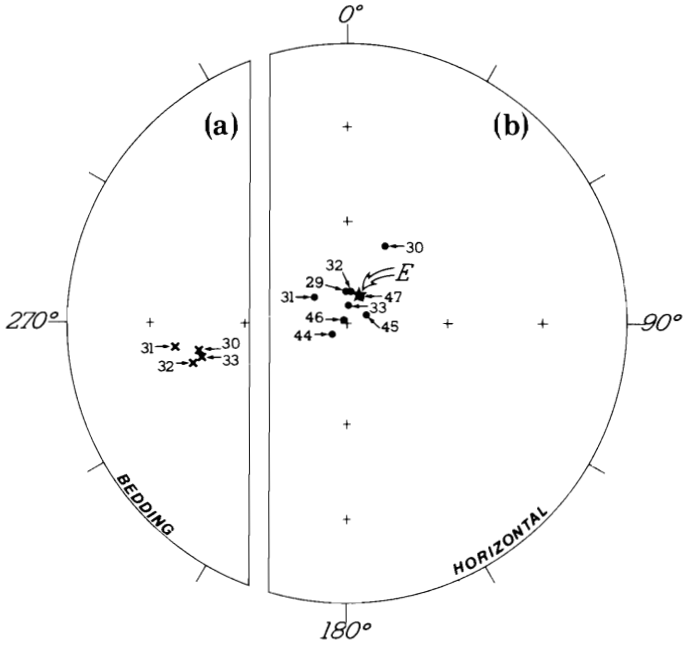


FIG. 10. (*a*) K_B and (*b*) K_C site directions.

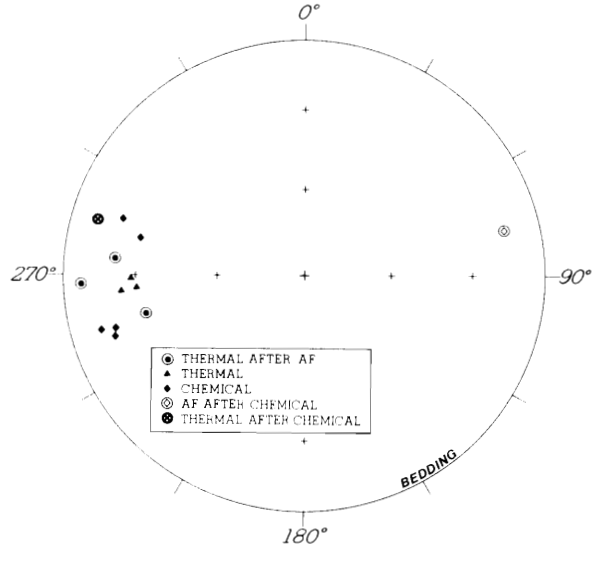


FIG. 11. K_A specimen directions. Closed symbols represent downward positive directions, and open symbols represent upward directions.

magnetization in the Big Spruce Complex (Irving and McGlynn 1976), which "could be caused by incipient weathering or by some as yet unstudied process related to the effects of ice-loading or permafrost." Some of the defined K_C may relate to the PEF; however, the immediately underlying Tsezotene Formation has a similar magnetization direction that is significantly different from the PEF. In fact, each of the six treatment mean directions of that study is significantly different from the PEF direction (Park and Aitken 1986). Therefore, we think that K_C is largely a secondary magnetization postdating the Paleocene folding but not of recent origin.

K_C also seems to be carried by magnetite, probably in the mafic silicate grains. One explanation for its presence in magnetite is that it represents a remagnetization, a partial

TABLE 7. Summary of K_A directions and poles

Description	N	Directions $D°, I°$	R	k	$\alpha_{95}°$	Poles °N, °W	$\delta_p°, \delta_m°$
1. Keele River, bedding	9	265, +19	8.80	40	8	07, 209	5, 9
2. Loretta Canyon, bedding	4	271, +25	3.87	22	20	12, 213	12, 22
3. Total, *in situ*	13	266, +17	12.63	33	7	06, 210	4, 8
4. Total, bedding	13	267, +21	12.64	33	7	09, 210	4, 8

NOTES: Symbols are explained in Tables 1, 2, and 4. Entry 4 is considered to be the best estimate of K_A.

TABLE 8. Application of tilt test

	N	Precision (k) of averages k_1 (hor)	k_2 (bdg)	k_2/k_1	95% confidence limit*	Significance of test
K_A	13 sp	32.63	33.45	1.05	1.98	No
K_B	36 sp	42.24	62.98	1.49	1.51	No
	4 si	63.03	326.47	5.18	4.28	Yes
K_C	58 sp	30.59	22.66	(1.35)	1.37	(No)
All sites	9 si	76.63	52.53	(1.46)	2.33	(No)
Keele sites	5 si	67.01	224.56	3.35	3.44	No
					2.59†	Yes
Loretta sites	4 si	153.67	138.18	(1.11)	4.28	(No)

NOTES: k is given before (hor) and after (bdg) correction for bedding; N is the number of sites (si) or specimens (sp). Parentheses refer to the ratio, k_1/k_2.
*Values are from McElhinny (1964).
†90% confidence limit.

viscous thermoremanent magnetization (pVTRM) caused by heating as a result of burial. To reset magnetite magnetizations with blocking temperatures around 550°C, theoretical temperatures around 450°C would be required (Pullaiah *et al.* 1975). Experimental evidence is somewhat at variance with theory and indicates much lower temperatures (e.g., 350°C over about 400 Ma) for resetting blocking temperatures of about 550°C (Pullaiah *et al.* 1975; Dunlop 1983). According to geological evidence, there was no deep burial during the Tertiary, and temperatures remained below those necessary for the onset of greenschist-facies metamorphism (300−375°C) (Turner 1968; McMechan and Price 1982; Thompson and Frey 1984). Nevertheless, temperatures may have been sufficient to reset much of the magnetite component. Another possible explanation for K_C in magnetite is that it was acquired as a CRM as the pigmental hematite altered to magnetite under reducing conditions. However, there is no evidence, as would be expected, for the reduction of the detrital hematite to magnetite.

Sequence of magnetization

K_B and one phase of K_C represent two pigmentation events, with K_B being prefolding and K_C mainly postfolding. Both phases of K_C appear to be at least partly postfolding, as indicated by a fold test applied to similar magnetizations in the underlying Tsezotene Formation (Park and Aitken 1986). However, results from the combined K_C phases at Keele River indicate a prefolding magnetization. Since the characteristics of the phases are similar in both the Tsezotene and Katherine units, we suggest that both pre- and postfolding K_C relates to the Laramide Orogeny.

K_A is associated with the detrital grains. The fact that there is no known prefolding event that would have removed an original magnetization residing in hematite strongly suggests that K_A is primary. Therefore the sequence of magnetization is K_A, K_B, and K_C.

Paleomagnetic apparent polar wander path

Poles associated with K_A, K_B, and K_C (Tables 4, 5, 7) are plotted in Fig. 12. All poles derived from Precambrian rocks of the Mackenzie Mountains are also plotted (circles and letters), together with Grenville (triangles) and other germane North American (squares) poles of the late Precambrian.

Primary and most secondary poles available from Proterozoic rocks of the Mackenzie Mountains are significantly different from most Hadrynian poles of the Grenville Province (Fig. 12*b*). We suggest that these Mackenzie Mountains poles lie on a separate APW path and that they define APW for a later Precambrian time than do those poles defining the Grenville Loop. Poles of the Grenville Track have recently been calibrated between 0.98 and 0.82 Ga (Berger *et al.* 1979) and between 0.98 and 0.88 Ga (Dallmeyer and Sutter 1980). The latter age spread is preferred here in view of the apparent timing of later events and assumed rates of plate movement (Fig. 12*b;* see below). Ages of the Mackenzie Mountains poles are known within broad limits. The oldest poles from the region are from the Mackenzie Mountains supergroup, whose maximum and minimum ages have been previously taken to be 1.2 and 0.7 Ga (Young *et al.* 1979), but all the suggested primary poles from the supergroup lie west of Grenville and older poles from North America and may belong to a younger time frame.

This separation of Grenville and Mackenzie Mountains poles could be postulated as the result of (1) the effect of relative rotations within the Mackenzie Mountains fold belt or between the fold belt and the craton; (2) the latter poles repre-

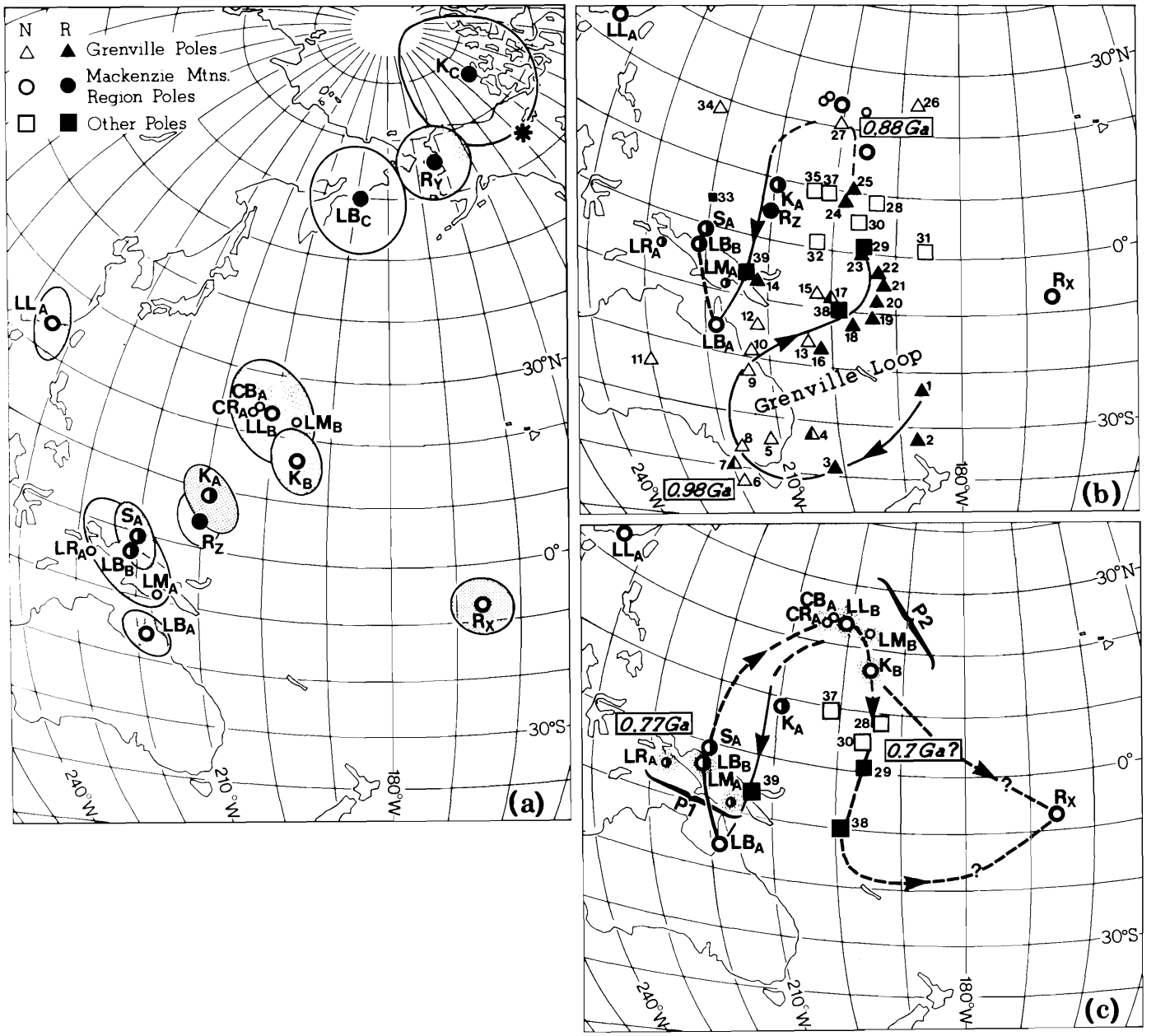


FIG. 12. Paleomagnetic poles from predominantly late Precambrian rocks of North America (1000−700 Ma). (*a*) Poles derived from the Mackenzie Mountains region together with their errors (δ_p, δ_m; or A_{95}). (*b*) Most relevant poles from North America showing the Grenville Loop and a proposed APW path drawn through poles of the Mackenzie Mountains region. (*c*) Proposed new polar path drawn through late Precambrian Mackenzie Mountains poles and other possibly relevant poles. An asterisk denotes the sampling region. Open symbols denote normal poles, and closed symbols denote reverse. Small symbols denote limited data. Mackenzie Mountains poles are labeled with letters. A subscript A denotes an interpreted primary pole; subscripts B and C denote secondary poles. Shaded poles are derived from magnetizations carried by red hematite pigment. In several units of the present study the pigment is probably primary. P1 and P2 denote general pigment directions. The poles shown are as follows. *Mackenzie Mountains poles:* CB_A, Basal Sandy member (Copper cycle) (J. K. Park, unpublished data); CR_A, Redstone River Formation (Copper cycle) (J. K. Park, unpublished data); K_A, K_B, K_C, Katherine Group (J. K. Park, this paper); LB_A, LB_B, LB_C, Basinal assemblage (Little Dal Group) (Park 1981*a*); LL_A, LL_B, lavas (Little Dal Group) (Morris and Park 1981; Morris and Aitken 1982); LM, LM_B, Mudcracked formation (Little Dal Group) (Park 1984*a*); LR_A, Rusty Shale formation (Little Dal Group) (J. K. Park, unpublished data); R_X, R_Y, R_Z, Rapitan Group (Morris 1977); S_A, sills in Tsezotene Formation (Morris and Park 1981; Park 1981*b*). *Grenville Structural Province poles:* 1, Mealy Mountain dykes, A component (Park and Emslie 1983); 2, 3, Thanet gabbro A_2 and A_1 components (Buchan 1978); 4, Grenville diorites, Quebec (322) (Irving *et al.* 1972); 5, Mealy Mountains dykes, B component (Park and Emslie 1983); 6, Morin anorthosite complex (482) (Irving *et al.* 1974*a*); 7, Allard Lake anorthosite (Hargraves and Burt 1967); 8, Haliburton intrusions, Hb_1 (Buchan and Dunlop 1976); 9, Whitestone diorite, WZ (503) (Ueno *et al.* 1975); 10, Lac St-Jean anorthosite, normal (Buchan *et al.* 1983); 11, Magnetawan metasediments (McWilliams and Dunlop 1978); 12, Wilberforce pyroxenite (457) (Palmer and Carmichael 1973); 13, Whitestone anorthosite, WW (500) (Ueno *et al.* 1975); 14, St-Urbain anorthosite (409) (P. B. Robertson and Roy 1979); 15, Indian Head anorthosite (Murthy and Rao 1976); 16, Seal Lake Group, B (Roy and Fahrig 1973; Park and Roy 1979); 17, Umfraville gabbro (Palmer *et al.* 1979); 18, Frontenac Axis dykes (363) (Park and Irving 1972); 19, Umfraville gabbro (Symons 1978); 20, Lac St-Jean anorthosite, reverse (Buchan *et al.* 1983); 21, Whitestone anorthosite, WY (502) (Ueno *et al.* 1975); 22, Haliburton intrusions, Hb_2 (Buchan and Dunlop 1976); 23, Morin anorthosite complex (483) (Irving *et al.* 1974*a*); 24, Grenville Front anorthosites (459) (Palmer and Carmichael 1973); 25, Michael gabbro (352) (Fahrig and Larochelle 1972); 26, Haliburton intrusions,

senting correlative poles from a separate landmass; or (3) the latter poles being younger. Relative rotations are commonly found within fold and thrust belts but may be difficult to detect by geological methods. There are two types of rotation to be considered: relative rotations between thrust sheets within the fold belt and relative rotation between the belt and the craton. Paleomagnetic poles may be used for detecting relative rotations in both cases. In the Mackenzie Mountains, pole evidence indicates that there has been little if any relative rotation between thrust sheets. K_A poles from the Loretta Canyon and Keele River localities agree within error (Table 1), though the regions are about 100 km apart and separated by at least one thrust fault and several major folds. Likewise, hematite pigment poles (CR_A, CB_A, LL_B, LM_B), from units within at least three different structural blocks, agree well (Fig. 12). Notably, pole LL_B from the Little Dal lavas in a region southwest of the Keele River locality is separated from the latter by at least two thrust faults. Other evidence is available from the Tsezotene Formation (Park and Aitken 1986). Evidence that the belt has not rotated relative to the craton is given below in the context of another discussion.

Both hypotheses (2) and (3) for explaining the position of the Mackenzie Mountains poles were earlier invoked to explain the seemingly unique position of the Grenville poles in relation to the distribution of North American poles (e.g., Irving *et al.* 1974*b*). If according to hypothesis (2) the Mackenzie Mountains poles are considered to be synchronous with the Grenville poles, the subparallel relationship of the two paths does not depict the classic Wilson cycle of converging landmasses (Wilson 1966; Irving *et al.* 1974*b*) but rather the contrary. Also, the sequence of poles from the Mackenzie Mountains supergroup is wrong for representing an APW path convergent with the Grenville Loop. Furthermore, regional geology as known and mapped (Aitken *et al.* 1982, Figs. 24, 33) denies any hypothesis of convergence of separate landmasses since 1.2 Ga (at least at the western continental margin), and the lower part of the Mackenzie Mountains supergroup can be correlated, largely on a formation-by-formation basis, with equivalents rooted in the Canadian Shield, as in the Brock Inlier and the Coppermine Homocline (Young 1977; Aitken 1982).

Hypothesis (3), that the Mackenzie Mountains poles are younger than poles of the Grenville Track, poses fewer problems. Morris and Park (1981) earlier postulated a younger, but antipolar, path from fewer data. The oldest pole so far determined from the Mackenzie Mountains is K_A of the Katherine Group; this primary pole in the APW path preferred here lies closest to the top of the Grenville Track (Fig. 12 *b* and *c*). A younger primary pole (LB_A), from the overlying Little Dal Group (Basinal assemblage), lies farther south, rather than north as called for by convergence. The track defined by the poles K_A and LB_A, soon to be supplemented by the primary pole from the Tsezotene Formation beneath the Katherine (Park and Aitken 1986) constitutes the most convincing polar wander sequence from the Mackenzie Mountains. However, this segment is undated. Its age is at least older than the 0.77 Ga date obtained from the diabases that intrude the Mackenzie Mountains supergroup as high as the lower Little Dal (Armstrong *et al.* 1982). A pole (S_A) from diabase sills intruding the Tsezotene Formation is close to hematite pigment poles (LM_A, LR_A, LB_B) from the Little Dal. This pigment direction, labelled P1, is the first of three from hematite pigments found in the Mackenzie Mountains supergroup and overlying Copper cycle rocks. In the Mudcracked (pole LM_A) and Rusty Shale (pole LR_A) formations, P1 appears to reside in hematite of primary origin and is probably a primary depositional remanence. In the Basinal assemblage, P1 has a clearly secondary, though unexplained, origin; there is a distinct difference between the primary pole LB_A and the pigment pole LB_B. The proximity of the P1 poles and the sill pole S_A indicates that the intrusion of the sills may be related to the event that ended deposition within the Mackenzie Mountains supergroup. Thus, the Mackenzie Mountains supergroup is apparently confined within the period 0.88 (youngest Grenville) to 0.77 Ga, and its paleomagnetic history may be represented by poles K_A, LB_A, LB_B, and S_A in sequence from oldest to youngest.

There are other scenarios for explaining pole S_A, but these would squeeze deposition of the Mackenzie Mountains supergroup into a narrower time frame. These scenarios are based on the fact that the sill pole S_A is widely divergent from the lava pole LL_A (Morris and Aitken 1982). Sills and lavas have been assumed by several authors to be of the same age but have never been demonstrated to be so. The data can be harmonized to a degree under the hypothesis that the lavas, whatever their age, are the older and that the sills (and the dykes of similar composition cutting the Little Dal) were emplaced during rifting at the inception of Copper cycle – Rapitan deposition. This hypothesis, however, leaves unexplained the close association of the S_A and Little Dal poles. According to present logic, pole LL_A should lie between primary pole LR_A from the Little Dal Group and probable primary poles CR_A and CB_A from lower units in the overlying Copper cycle rocks. Clearly it does not; therefore we reject LL_A as a primary pole.

Several other poles from the North American craton may fall on the APW path of the Mackenzie Mountains supergroup, but none of these is dated. One pole (39) from the Reynolds Point Formation of Victoria Island lies between the Katherine Group (K_A) and Basinal assemblage (LB_A) poles. The Reynolds Point Formation is part of the autochthonous, Proterozoic, epicratonic, sedimentary cover of Victoria Island. Its correlation with the lower Little Dal Group of the Mackenzie Mountains, 500 km to the southwest (Aitken *et al.* 1978; Young 1979), remains unchallenged. Because that correlation depends on position in a sequence, position between two correlatable evaporitic units, and support from stromatolite biostratigraphy, it cannot be reviewed within the compass of this paper; the interested reader is referred to the sources cited. The reasonably close association between the Reynolds Point and lower Little Dal poles also argues against any significant relative rotation between the Mackenzie Mountains and Victoria Island sites

Hb_3 (Buchan and Dunlop 1976); 27, Thanet gabbro, B (Buchan 1978); 34, Tudor gabbro (458) (Palmer and Carmichael 1973). *Other North American poles:* 28, Franklin diabases, normal (27 sites) (Fahrig *et al.* 1971; Fahrig and Schwarz 1973; Park 1974; Christie and Fahrig 1983); 29, Franklin diabases, reverse (51 sites) (Fahrig *et al.* 1971; W. A. Robertson and Baragar 1972; Fahrig and Schwarz 1973; Park 1974, 1981*c*; Palmer and Hayatsu 1975; Christie and Fahrig 1983); 30, Spokane Formation, overprint (Vitorello and Van Der Voo 1977); 31, Rama diabase, Arizona (391) (Larson *et al.* 1973); 32, Uinta Mountain Group, lower sites (Bressler 1981); 33, Uinta Mountain Group, upper site (Bressler 1981); 35, 36, Mount Nelson Formation, secondary and primary(?) (Morris and McMechan 1983); 37, 38, Natkusiak basalts, normal and reverse (Palmer and Hayatsu 1975; Palmer *et al.* 1983); 39, Reynolds Point Formation (Palmer *et al.* 1983). Numbers in parentheses above refer to the Ottawa pole catalogue list (Irving and Hastie 1975).

and, the latter being autochtonous and epicratonic, against relative rotation betwen the Mackenzie Mountains sites and Laurentia (including Grenvillia). We also point out that the pole sequences of the Mackenzie Mountains and Grenville APW paths are oppositely directed and that these cannot be made to agree by any simple rotation.

Another rock unit, the Uinta Mountain Group of Utah and Colorado, has also been interpreted as correlative with the Little Dal Group (Bressler 1981), but apart from the proximity of poles, there is no independent evidence for a correlation. A pole (33) from the highest site sampled in the unit is markedly different from a pole (32) calculated from the lower five sites. Since no chemical work was undertaken to separate the magnetizations of the red pigment and detrital grains, it is not possible to assess the reliability of these poles; they could represent composite magnetizations.

Poles 14 and especially 11 and 34, from the Grenville terrane, lie far off the Grenville Track, as depicted in Fig. 12*b*. Poles 14 and 11 are thought to represent stages in the cooling history of the Grenville Province, as do most other Grenville poles so far obtained. Pole 14 is from the St-Urbaine anorthosite (P. B. Robertson and Roy 1979), which is thought to be slightly older than transecting dykes dated at 890 Ma (Rose 1961). Pole 11 from iron-rich metasediments near Magnetawan, Ontario, is thought to be a so-called Grenville "A" pole derived from a magnetization carried by magnetite and hematite (McWilliams and Dunlop 1975). It lies far off the Grenville Loop as defined in Fig. 12, but no obvious argument relates it to the proposed polar path. Pole 34 is from the Tudor gabbro (Palmer and Carmichael 1973). It is thought to be associated with an age of 670 Ma (K−Ar isochron, Hayatsu and Palmer 1975), and therefore may lie on a younger APW path, such as the path inferred (below) between LR_A (or S_A) and CR_A.

The route of APW following the path from the Mackenzie Mountains supergroup is more speculative. As a possible route, we suggest a direct connection with further poles (CR_A, CB_A, LL_B, LM_B, K_B) from the Mackenzie Mountains (Fig. 12*c*). These secondary poles document a pervasive red hematite magnetization, which is found in most units of the Mackenzie Mountains supergroup and the Copper cycle (Park 1984*b*) but apparently not the overlying Rapitan Group (Sayunei Formation). The magnetization is probably primary in rocks of the Copper cycle (CR_A, CB_A) but secondary in units of the Mackenzie Mountains supergroup (LL_B, LM_B, K_B). We call this hematite pigment and its direction P2 to distinguish it from the apparently earlier pigmentation P1 confined to the Little Dal Group. One site in the Little Dal Mudcracked formation carries both P1 and P2 (Park 1984*a*). Because the Copper cycle and the Rapitan are partly conformable, it is likely that P2 relates closely to the beginning of the rifting that initiated the Copper cycle and that continued into early Rapitan (Sayunei) time and possibly into middle Rapitan (Shezal) time. The large break between LR_A or S_A and P2 possibly documents a long hiatus occurring prior to the beginning of Copper cycle deposition. Following P2, the APW path may either continue on to the primary(?) R_X pole of the Sayunei Formation or pass through the Franklin poles (28, 29) and then perhaps on to R_X.

It is tempting to connect the P2 poles to the nearby Franklin poles because of the possible age and age trend within the Franklin. Age data for the Franklin diabases are presently limited to K−Ar determinations but do establish a minimum age of about 675 Ma (Fahrig *et al.* 1971). The Natkusiak volcanics of Victoria Island, which are probably related to the Franklin diabases (Palmer and Hyatsu 1975), are above units that have been equated to the Copper cycle and possibly younger rocks above the Mackenzie Mountains supergroup (Young 1977; Jefferson *et al.* 1985). If the diabases that intrude the latter sequence as high as the lower Little Dal Group also give a terminal pole for that sequence, as appears likely from the paleomagnetic data, then the Franklin diabases are younger. They would be younger than 770 Ma and thus confined to the period 770−675 Ma. Normal (28) and reverse (29) poles from the Franklin, which are significantly different, define a possible age sequence from north to south. This trend, perhaps significantly, follows directly from the P2 poles of the Mackenzie Mountains. The apparent age sequence of normal to reverse Franklin diabase poles is apparently supported by the sequence of normal (37) to reverse (38) poles from the associated Natkusiak basalts of Victoria Island (Palmer *et al.* 1983).

The younger APW path might also include pole 26 (Buchan and Dunlop 1976) from the Haliburton intrusions and pole 27 (Buchan 1978) from the Thanet gabbro. Both have been traditionally assigned to the top of the Grenville Track. They are each thought to be the youngest of three poles calculated from their respective units. Pole 26 has been interpreted from $^{40}Ar/^{39}Ar$ dating to be ≤ 820 Ma (Berger *et al.* 1979), which is younger than the age assigned by Dallmeyer and Sutter (1980) to the top of the Grenville Track.

Other poles determined from the Precambrian rocks of the Mackenzie Mountains (LB_C, K_C, R_Y), apart from R_Z of the Sayunei Formation, are in high latitudes. They may relate to the Laramide Orogeny and later events or to an earlier cycle. Pole LB_C from the Little Dal Basinal assemblage is definitely prefolding (Park 1981*a*). Pole R_Y from hematite pigment in the Rapitan Group is thought to be prefolding and possibly related to late Precambrian glaciation, though it could also be interpreted as a recent postfolding pole (Morris 1977). Pole K_C from the present study may in part be prefolding. The question is whether the prefolding magnetizations slightly predate the Laramide Orogeny or relate to a much earlier period, such as the end of the Precambrian. The R_Y pole uncorrected for tilt (estimated at 73°N, 116°W) is very close to the K_C pole (77°N, 122°W), and both poles relate to a pigmentation event. LB_C would fit into a pigmentation event. The characteristics of its magnetization, though earlier ascribed to goethite (Park 1981*a*), would relate better to the hydrogoethites (α-$FeOOH \cdot nH_2O$); both minerals, though, are precursors of hematite pigment (Bagin *et at.* 1978). Therefore, we suggest that all three poles represent a pigmentation event P3, which occurred just prior to, during, and immediately following the Laramide Orogeny.

The APW path just described neatly incorporates most of the paleomagnetic data from the Mackenzie Mountains and simply relates them to the geological history. There is no evidence that the configuration of the path is an artifact of relative rotations between structural blocks or that the path itself is relatively shifted because of relative rotation between fold belt and craton. We emphasize, however, that the latter interpretation presently rests on the reasonable agreement between two poles of the geologically correlated Reynolds Point Formation from Victoria Island on the craton and the lower Little Dal Group of the Mackenzie Mountains. Obviously more data are needed to firm up this conclusion.

Paleolatitudes

The paleomagnetic study indicates that the craton in the sampling region was at 11°N (or 11°S) during the formation of the Katherine Group, at 15°S (or 15°N) during the formation of the overlying lower Little Dal Group, and at the equator during the intrusion of the sills. Over the whole period, the craton was oriented at about right angles to its present position relative to geographic north (93, 95, and 90° clockwise, respectively, for the three units).

Summary

(1) The Katherine Group yielded three paleomagnetic poles: one primary and two secondary. The primary magnetization probably resides in detrital hematite; the secondary magnetization resides mainly in the red pigment. One of the secondary magnetizations is largely postfolding (post-Paleocene) and is found in both the pigment and in possible magnetite in mafic silicate particles. The other relates to a pervasive hematite pigmentation event of the Precambrian.

(2) During the formation of the Katherine Group, the craton was at 11°N (or 11°S) and rotated by 93° clockwise (or 87° anticlockwise) relative to the present geographic north.

(3) Poles from the Mackenzie Mountains supergroup form an APW path, which we suggest is separate from and younger than the Grenville Loop and dates from the period between 0.88 and 0.77 Ga. Present evidence indicates that none of the constituent poles have been rotated relative to one another or to poles of the craton.

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